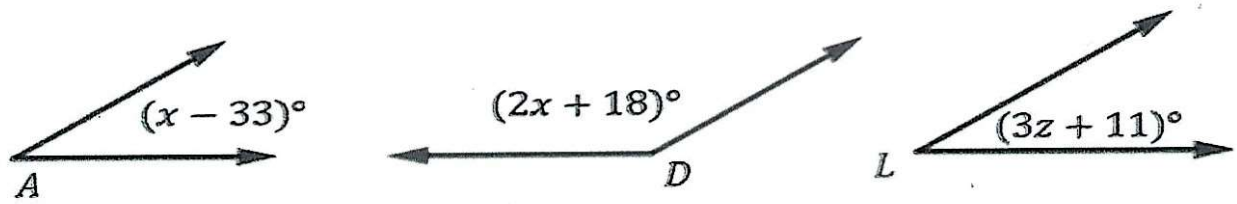
Unit 3: Worked Examples

1. Given $\angle A$ is supplemental to $\angle D$ and $\angle L$ is supplementary $\angle D$, determine the value of x.



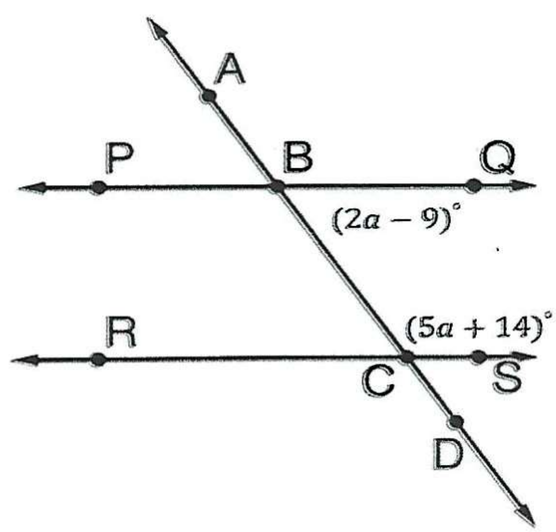
$x - 33 + 2x + 18 = 180$

$3x - 15 = 180$

$3x = 195$

$x = 65$

2. $\overleftrightarrow{PQ} \parallel \overleftrightarrow{RS}$ where $\overleftrightarrow{AD}$ is a transversal. Determine the value of a.



$2a - 9 + 5a + 14 = 180$

$7a + 5 = 180$

$7a = 175$

$a = 25$